POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2026
Programme: BE | Full Marks : 100
Course: Image Processing and Pattern Recognition (New) | Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Discuss the concepts of image sampling and quantization. Given two distinct pixels **p** and **q**, explain under what conditions m-adjacency is preferred over 8-adjacency to prevent ambiguities in path connectivity. 8

   b) Define Image Enhancement in spatial domain? List and explain any two gray level transformation Technique. 7

2. a) Explain how histogram equalization contributes to the image enhancement. Perform the histogram equalization on the following 8-bit gray level image. 8

3	2	4	5
1	2	2	2
7	3	1	2
7	6	4	5

   b) Define DCT Transform with its expressions? List out its properties and use cases. 7

3. a) Derive the Kernel for Haar Transform of N = 2 order. 7

OR

Perform Haar transform on following image and retain the original image after transformation.

1	5	3
9	7	2
6	8	4

   b) What is image restoration? List down different noise models in an image with the graphs and equation. 8

4. a) Construct Huffman code for each gray level given and calculate the average code length, source entropy, compression ratio and efficiency of the given data. 8

Rk	0	1	2	3	4	5	6	7
Nk	30	45	50	60	15	10	35	90

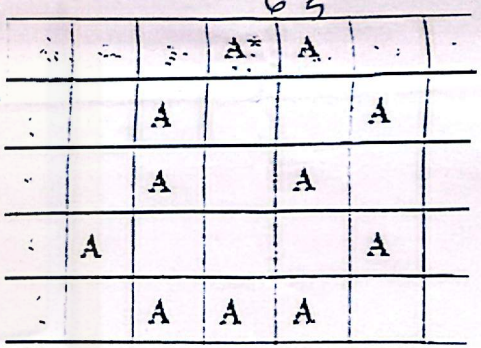
   b) Why segmentation is needed in image processing? Describe region growing and merging with an example. 7

5. a) Explain how global thresholding, adaptive thresholding, and multi-level thresholding separate foreground objects from complex backgrounds. 8

   b) What is line detection? Explain the line detection with a suitable detailed example. 7

OR

Find the chain code and shape number of the given pixel relationships. A* is the starting point and rotation is clockwise direction.



6. a) Define PCA. Explain its step to illustrate how feature from an image is extracted? 8

   b) What is an Artificial Neural Network (ANN) in the context of pattern recognition? Explain the architecture and mathematical operation of a single-layer perceptron. 7

7. Write short notes on: **(Any two)** 2×5

   a) LZW coding

   b) Signature

   c) SIFT

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